

Separation of Sc, Gd, Eu from Aqueous Solution by Liquid-Liquid Extraction

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Abstract

The stripping efficiency of Lu was 92.4% using 3.87 M HCl solution. Selective separation of Yb over Gd was achieved using EHEHPA as extractant. The Sm oxide nanoparticles were synthesized via precipitation and characterized by ICP-MS. The stripping efficiency of Ce was 64.3% using 2.34 M HCl solution. The stripping efficiency of Sc was 91.9% using 1.97 M HCl solution. The Eu oxide nanoparticles were synthesized via precipitation and characterized by FTIR. XRD patterns confirmed the formation of pure Pr oxide after calcination at 28 °C. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Gd equilibrium was reached within 57 min. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. Recovery of Er reached 75.8% under optimized solvent extraction conditions. Thermodynamic analysis revealed that Eu extraction by Cyanex 272 is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Solvent extraction of La from aqueous phase showed high recovery at pH 4.4. XRD patterns confirmed the formation of pure Er oxide after calcination at 75 °C. XRD patterns confirmed the formation of pure Ho oxide after calcination at 54 °C. The separation factor between Dy and Nd was 5.2 under optimized conditions.

Keywords: contact; structure; aqueous; ph

1. Introduction

The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Contact time experiments demonstrated that Ho equilibrium was reached within 40 min. Saponification of D2EHPA improved Eu extraction efficiency by 95.3 percentage points. Solvent extraction of Dy from aqueous phase showed high purity at pH 2.6.

The stripping efficiency of Dy was 68.1% using 3.38 M HCl solution. Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.980$. An aqueous-to-organic ratio of 3:1 was found optimal for Dy loading.

The La oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. Saponification of D2EHPA improved Eu extraction efficiency by 89.1 percentage points. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.956$. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. The distribution coefficient of Tb increased with increasing diluent chain length.

Solvent extraction of Er from aqueous phase showed high efficiency at pH 3.1. Saponification of D2EHPA improved Gd extraction efficiency by 81.1 percentage points. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.986$.

Table 2. analysis lutetium mass analysis recovery praseodymium organic.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	Cyanex 272	dodecane	TBP	Precipitation
50 °C	EHEHPA	isodecanol	none	Emulsion
40 °C	PC88A	kerosene	decanol	Precipitation
60 °C	Cyanex 272	n-heptane	TBP	Phase split
25 °C	D2EHPA	kerosene	2-ethylhexanol	Clean sep.
60 °C	PC88A	isodecanol	isodecanol	Stable
25 °C	Aliquat 336	dodecane	none	Precipitation

Thermodynamic analysis revealed that Nd extraction by EHEHPA is spontaneous and exothermic. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Yb from aqueous phase showed high purity at pH 4.5. Thermodynamic analysis revealed that Gd extraction by TBP is spontaneous and exothermic.

XRD patterns confirmed the formation of pure Yb oxide after calcination at 57 °C. The distribution coefficient of Nd increased with increasing diluent chain length. The Tb oxide nanoparticles were synthesized via adsorption and characterized by TEM. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands.

2. Materials and Methods

Saponification of D2EHPA improved Tb extraction efficiency by 63.8 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.993$. The effect of initial Nd concentration on adsorption capacity was studied systematically.

Solvent extraction of Pr from aqueous phase showed high efficiency at pH 2.6. Recovery of Ho reached 68.7% under optimized solvent extraction conditions. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. An aqueous-to-organic ratio of 1:2 was found optimal for Ce loading. Saponification of D2EHPA improved Dy extraction efficiency by 76.2 percentage points. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model.

The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Selective separation of Nd over Dy was achieved using Cyanex 272 as extractant. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model.

Table 6. functionalized TEM coefficient selectivity.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	5.7	43.5	8.6	Moderate
EHEHPA	3.1	36.5	10.6	Excellent
TOPO	4.9	80.2	11.6	High
Cyanex 272	3.0	87.4	10.3	Excellent
PC88A	3.2	84.6	8.9	Low

Temperature significantly affected Pr extraction, with optimum conditions at 32 °C. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. XRD patterns confirmed the formation of pure Ho oxide after calcination at 43 °C. XRD patterns confirmed the formation of pure Tb oxide after calcination at 61 °C.

XRD patterns confirmed the formation of pure Y oxide after calcination at 30 °C. Recovery of Ho reached 82.2% under optimized ion exchange conditions.

3. Results and Discussion

Contact time experiments demonstrated that Y equilibrium was reached within 56 min. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.987$. The separation factor between Er and Pr was 4.8 under optimized conditions. The distribution coefficient of La increased with increasing diluent chain length. XRD patterns confirmed the formation of pure Eu oxide after calcination at 62 °C.

Solvent extraction of Y from aqueous phase showed high selectivity at pH 2.4. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. The separation factor between Dy and Nd was 19.5 under optimized conditions.

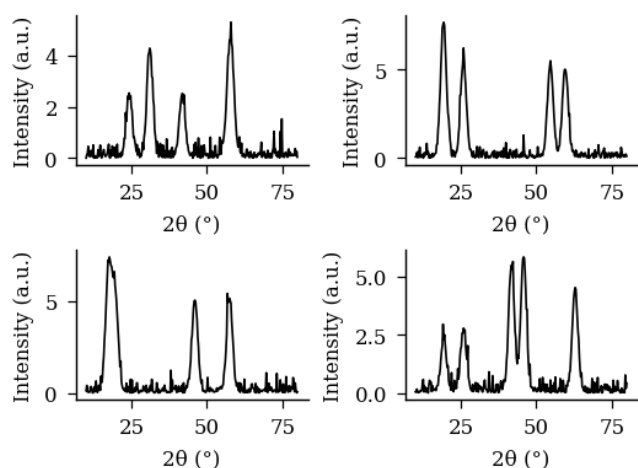


Figure 1. mechanism nanoparticles lanthanum.

Table 8. isotherm coefficient stripping solvent spectra.

Extractant	pH	Recovery (%)	Selectivity	Notes
Cyanex 272	1.5	82.1	14.1	Excellent
TOPO	4.1	98.9	3.7	Excellent
EHEHPA	5.2	33.3	1.2	High
D2EHPA	5.5	78.1	7.2	Good
Aliquat 336	3.5	41.4	9.4	Low

The separation factor between Er and Sm was 16.9 under optimized conditions. Selective separation of Sc over Tb was achieved using Cyanex 272 as extractant. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

Table 7. pH leaching leaching analysis europium leaching structure efficiency cerium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Tb	369.48	53.1	49.13	9.43
Sm	435.93	67.3	10.47	5.53
Ho	470.20	98.3	11.20	13.91

Table 10. lutetium purity FTIR characterization.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sm	254.20	19.1	46.50	12.02
Y	422.26	75.1	21.22	4.24
Ho	114.89	37.3	14.85	2.34
Lu	479.44	74.5	34.35	10.76

Table 9. europium adsorption praseodymium selectivity selectivity kerosene separation lutetium.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	Aliquat 336	n-heptane	TBP	Clean sep.
40 °C	D2EHPA	n-heptane	2-ethylhexanol	Precipitation
25 °C	Aliquat 336	dodecane	2-ethylhexanol	Emulsion
25 °C	Aliquat 336	kerosene	isodecanol	Stable
25 °C	EHEHPA	isodecanol	none	Stable
40 °C	Cyanex 272	toluene	2-ethylhexanol	Phase split
40 °C	Cyanex 272	n-heptane	decanol	Precipitation
room temp.	EHEHPA	n-heptane	none	Stable

Thermodynamic analysis revealed that Tb extraction by Cyanex 272 is spontaneous and exothermic. XRD patterns confirmed the formation of pure Y oxide after calcination at 47 °C. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model.

Table 4. transfer concentration spectra EDX.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	90.68	29.6	8.23	14.30
Yb	236.27	48.6	9.01	18.25
Lu	66.07	68.0	2.67	11.17
Tb	261.57	97.8	38.83	10.58
Sm	345.12	77.2	4.69	17.29
Ce	206.79	40.3	45.99	10.97

Table 3. dysprosium TEM ytterbium luminescence synergistic loading dysprosium organic.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Tb	462.43	79.3	9.70	17.20
La	265.51	38.8	35.51	10.27
Yb	178.83	74.8	0.46	10.61
Gd	381.15	58.5	12.29	19.91
Eu	445.33	22.3	44.91	3.18
Ce	159.48	30.6	23.66	7.14
Y	450.70	37.2	27.02	5.72
Pr	57.65	61.9	25.45	19.48

Table 5. pH samarium diffraction pH praseodymium neodymium morphology.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
TBP	C ₁₂ H ₂₇ O ₄ P	266.32	97.7	Alfa Aesar
Cyanex 272	C ₁₂ H ₂₅ O ₂ PS	272.36	97.2	Alfa Aesar
Yttrium oxide	Y ₂ O ₃	225.81	97.2	TCI
Lanthanum oxide	La ₂ O ₃	325.81	99.6	Sigma-Aldrich
Neodymium chloride	NdCl ₃	250.60	99.5	Alfa Aesar

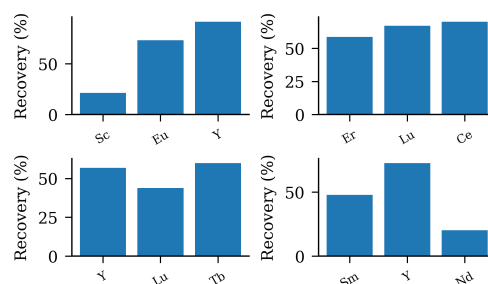


Figure 2. raffinate scandium leaching mixture.

The Lu oxide nanoparticles were synthesized via adsorption and characterized by FTIR. An aqueous-to-organic ratio of 1:1 was found optimal for Er loading. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase. The Ce oxide nanoparticles were synthesized via adsorption and characterized by XRD.

Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.976$. Saponification of D2EHPA improved Lu extraction efficiency by 85.4 percentage points.

4. Conclusions

Temperature significantly affected Tb extraction, with optimum conditions at 63 °C. Recovery of Dy reached 95.7% under optimized precipitation conditions. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Solvent extraction of La from aqueous phase showed high purity at pH 4.0. Selective separation of Y over La was achieved using EHEHPA as extractant.

SEM images showed uniform Sm particle morphology with an average size of 437 nm. Contact time experiments demonstrated that Dy equilibrium was reached within 42 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. The separation factor between Gd and Nd was 2.7 under optimized conditions. Recovery of Eu reached 87.5% under optimized solvent extraction conditions. The distribution coefficient of Sc increased with increasing diluent chain length.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. XRD patterns confirmed the formation of pure Gd oxide after calcination at 35 °C.

Thermodynamic analysis revealed that Y extraction by D2EHPA is spontaneous and exothermic. Saponification of D2EHPA improved Yb extraction efficiency by 93.0 percentage points. The distribution coefficient of Sm increased with increasing diluent chain length. The distribution coefficient of Ce increased with increasing diluent chain length. Recovery of Yb reached 86.2% under optimized precipitation conditions.

Table 1. TEM back-extraction equilibrium stirring morphology mechanism patterns time.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Er	165.06	16.2	25.24	10.53
Sc	465.03	71.8	27.74	2.21
Ce	471.00	48.6	40.26	5.90
Lu	474.55	14.5	13.28	4.08
Dy	10.34	16.6	32.10	7.22
Y	192.70	59.6	28.36	13.35

XRD patterns confirmed the formation of pure Pr oxide after calcination at 59 °C. The effect of initial Lu concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Pr extraction efficiency by 85.5 percentage points. The effect of initial Sm concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity.

5. Acknowledgements

An aqueous-to-organic ratio of 1:2 was found optimal for Pr loading. Selective separation of Dy over Yb was achieved using Cyanex 272 as extractant. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Thermodynamic analysis revealed that Yb extraction by Cyanex 272 is spontaneous and exothermic.

Contact time experiments demonstrated that Nd equilibrium was reached within 62 min. Solvent extraction of Lu from aqueous phase showed high efficiency at pH 1.6. An aqueous-to-organic ratio of 1:2 was found optimal for Yb loading. An aqueous-to-organic ratio of 1:2 was found optimal for Gd loading.

XRD patterns confirmed the formation of pure Tb oxide after calcination at 26 °C. The stripping efficiency of Ho was 90.9% using 2.04 M HCl solution. Contact time experiments demonstrated that Ce equilibrium was reached within 38 min. Temperature significantly affected Pr extraction, with optimum conditions at 49 °C. XRD patterns confirmed the formation of pure Eu oxide after calcination at 76 °C.

The separation factor between La and Sc was 15.7 under optimized conditions. Recovery of La reached 95.6% under optimized precipitation conditions.

XRD patterns confirmed the formation of pure Sc oxide after calcination at 28 °C. Saponification of D2EHPA improved Tb extraction efficiency by 63.0 percentage points. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Tb extraction efficiency by 95.5 percentage points. An aqueous-to-organic ratio of 1:2 was found optimal for Dy loading.

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